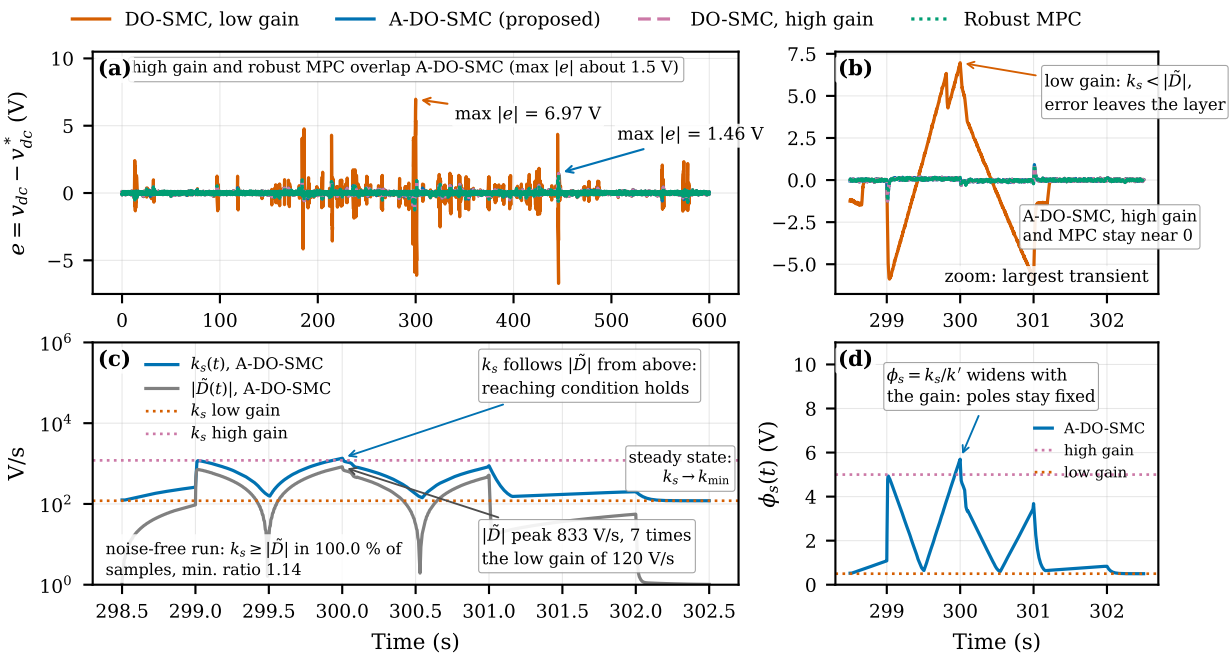

DO-SMC, low gain
A-DO-SMC (proposed)
DO-SMC, high gain
Robust MPC
(a)
high gain and robust MPC overlap A-DO-SMC (max |e| about 1.5 V)
max |e| = 6.97 V
max |e| = 1.46 V
e = v_dc − v*_dc (V)
(b)
low gain: k_s < |D̃|, error leaves the layer
A-DO-SMC, high gain and MPC stay near 0
zoom: largest transient
(c)
k_s(t), A-DO-SMC
|D̃(t)|, A-DO-SMC
k_s low gain
k_s high gain
k_s follows |D̃| from above: reaching condition holds
steady state: k_s → k_min
|D̃| peak 833 V/s, 7 times the low gain of 120 V/s
noise-free run: k_s ≥ |D̃| in 100.0 % of samples, min. ratio 1.14
V/s
Time (s)
(d)
φ_s = k_s/k′ widens with the gain: poles stay fixed
A-DO-SMC
high gain
low gain
φ_s(t) (V)
Time (s)